Publishing a data visualisation

EDC117: Week 7

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Sign in

Learning outcomes

What will I learn?

By the end of this week you will know:

- How to take the publication medium your data visualisation will feature in into account when exporting and deciding on size, orientation, and so on
- How to apply different technical terms from print & design to describe differences in quality, orientation, size (e.g. aspect ratio), and method of reproduction (rasterised versus vectorised)
- How to make your data visualisation look good no matter where it's published!

That's a nice data visualisation...

That's a nice data visualisation… How does it look on your phone?

That's a nice data visualisation… How does it look on your laptop?

That's a nice data visualisation… How does it look projected on the wall?

That's a nice data visualisation… How does it look projected in a print newspaper?

Rarely is the same data visualisation suitable for everywhere it might be seen.

You might need to customise your data visualisation for different mediums, or test how it looks on different publication formats.

Don't be the person that puts a screenshot of a dashboard in your report!

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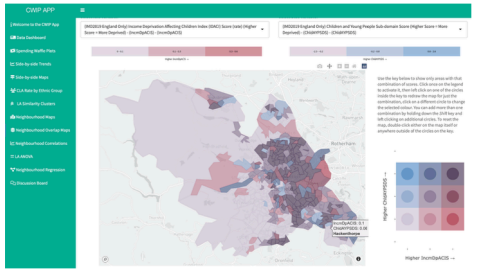


Fig.1 Vivamus semper, nisi non laoreet bibendum, orci tellus rutrum odio, quis pharetra velit purus vitae metus.

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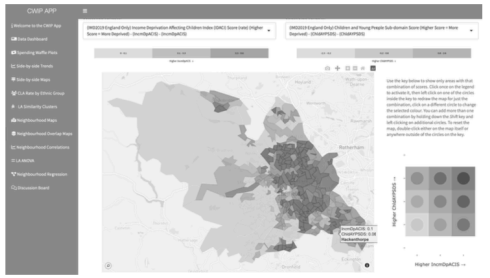


Fig.1 Vivamus semper, nisi non laoreet bibendum, orci tellus rutrum odio, quis pharetra velit purus vitae metus.

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Or worse!

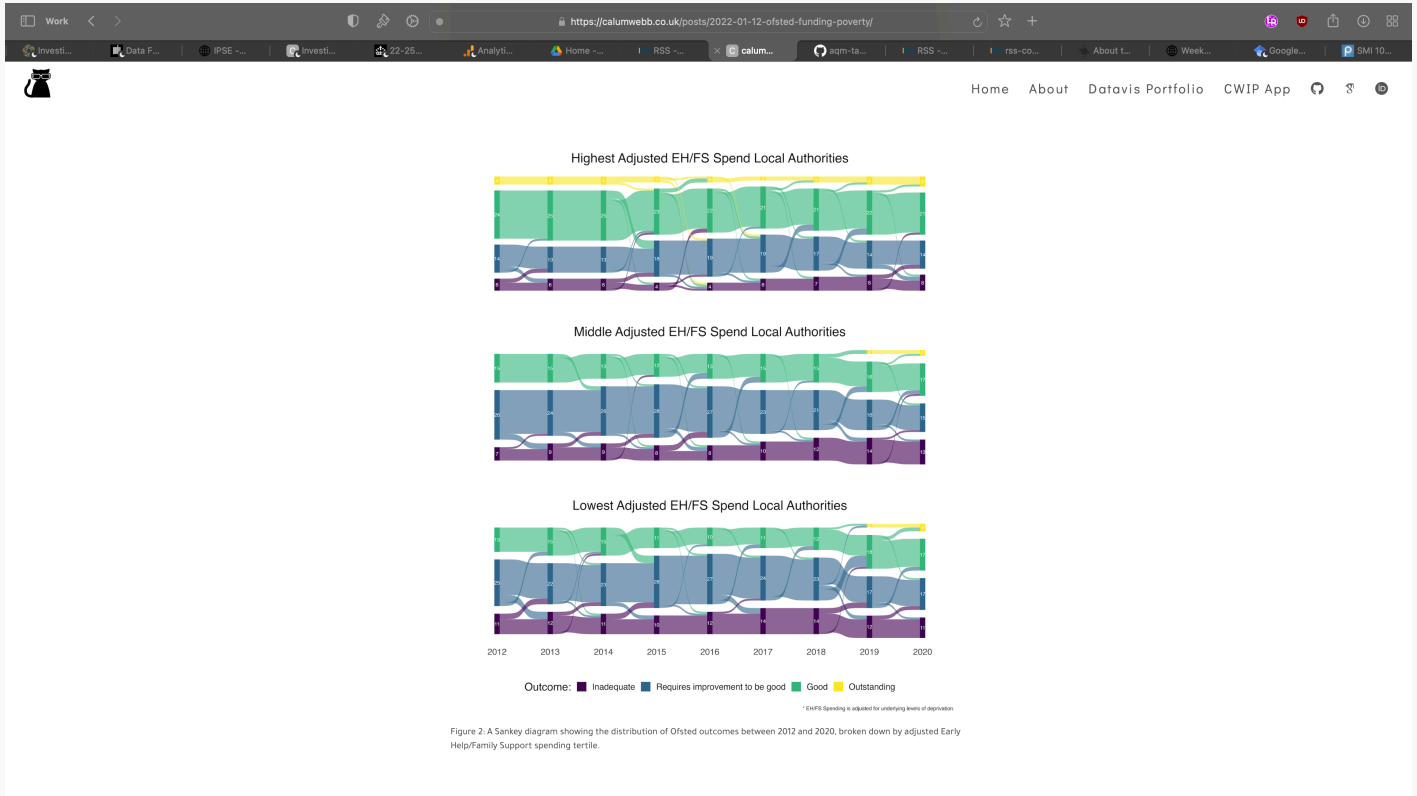


Figure 2: A Sankey diagram showing the distribution of Ofsted outcomes between 2012 and 2020, broken down by adjusted Early Help/Family Support spending tertile.

- An image of a Sankey plot from my personal website when viewed on a web browser...

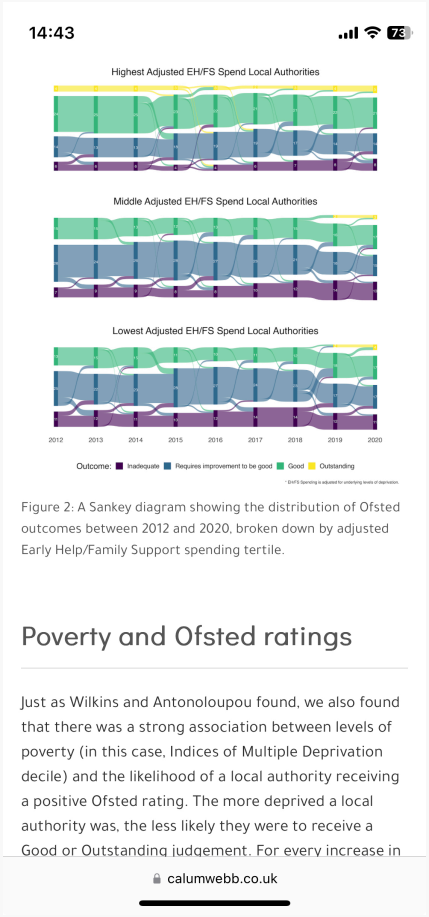


Figure 2: A Sankey diagram showing the distribution of Ofsted outcomes between 2012 and 2020, broken down by adjusted Early Help/Family Support spending tertile.

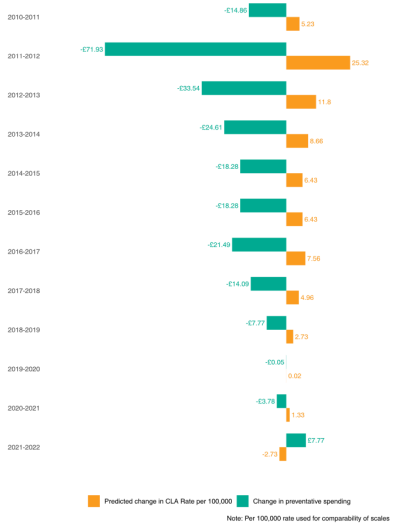
- An image of a Sankey plot from my personal website when viewed on a web browser...
- And on my mobile phone.

Luckily the same image works fairly well regardless, but maybe the text is a bit too small for phones.

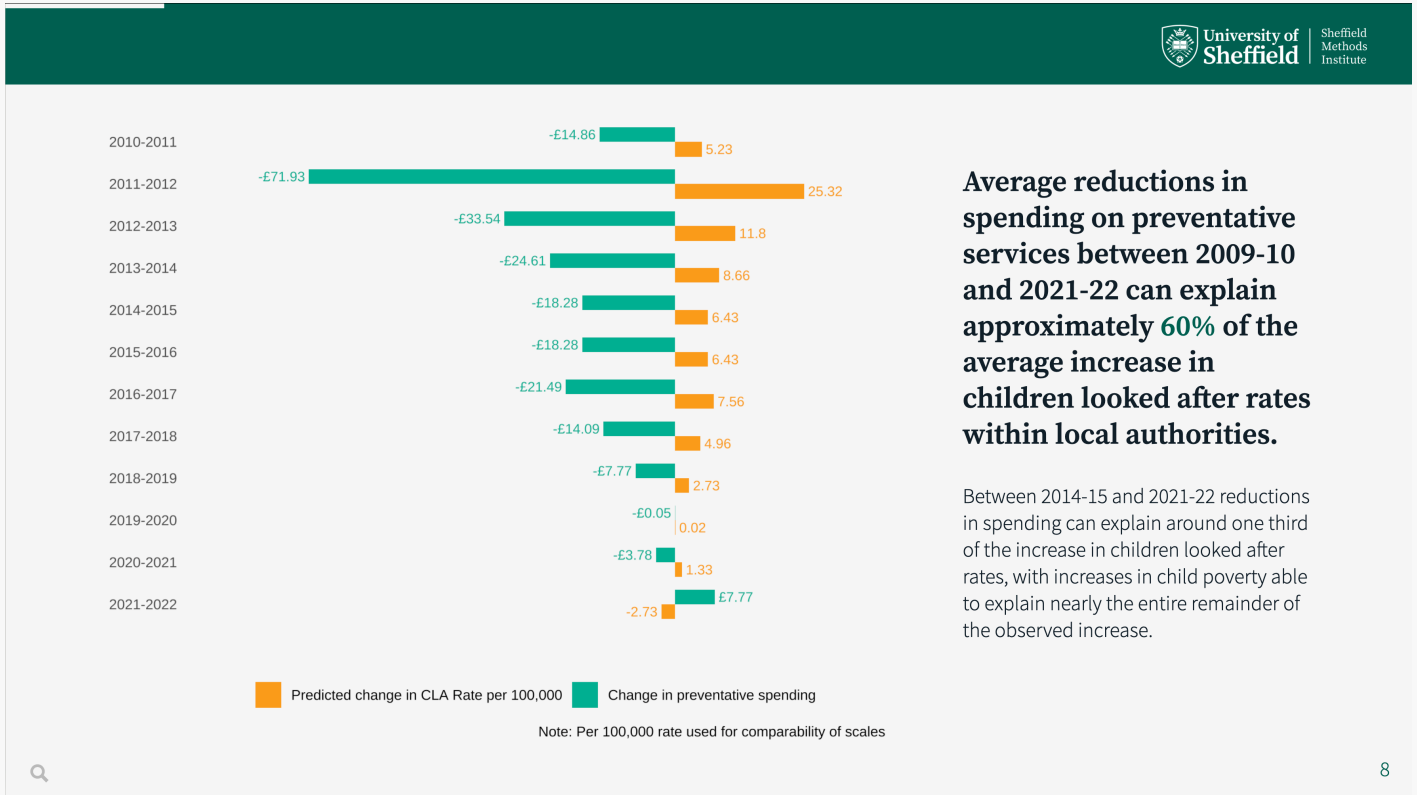
Does spending more on preventative services reduce rates of intervention?

In an average sized local authority of around 56,000 children this £100 per child increase per child would equate to an expected decrease of around 20 fewer children in care. A typical local authority has reduced their spending on preventative services by £212 per child when compared to 2009-10, implying around 42 more children in care over two years than we would otherwise expect has spending remained the same. Based on a child population estimate of around 11million children living in English local authorities, we would expect an estimated 8,300 more children in care nationally (around 10.1% of the children looked after population in 2021-22) — than might otherwise be the case had preventative spending been kept at 2009-10 levels.

Figure 3: Breakdown of changes in average preventative services spending changes between years and their predicted impact on children looked after rates over a two year interval (in 100,000s)



- Here's a graph that's from a research report I've written...



- Here's a graph that's from a research report I've written...
- But here's how it looks when I present it using a slidedeck...



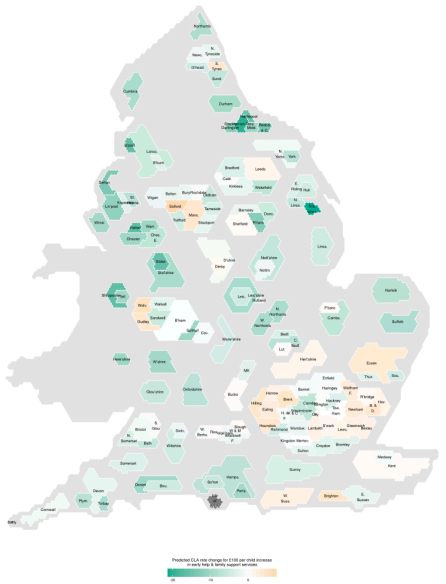
- Here's a graph that's from a research report I've written...
- But here's how it looks when I present it using a slidedeck...
- And here's how it looks on the website

I've changed the aspect ratio so it fits the space better (not trying to squeeze a vertical image on a horizontal space), and on the website changed the theme

background to blend in and make it look

When and where has preventative spending been most effective?

Figure 11: Cartogram of variation in spending effectiveness for reducing children looked after rates by local authority



89

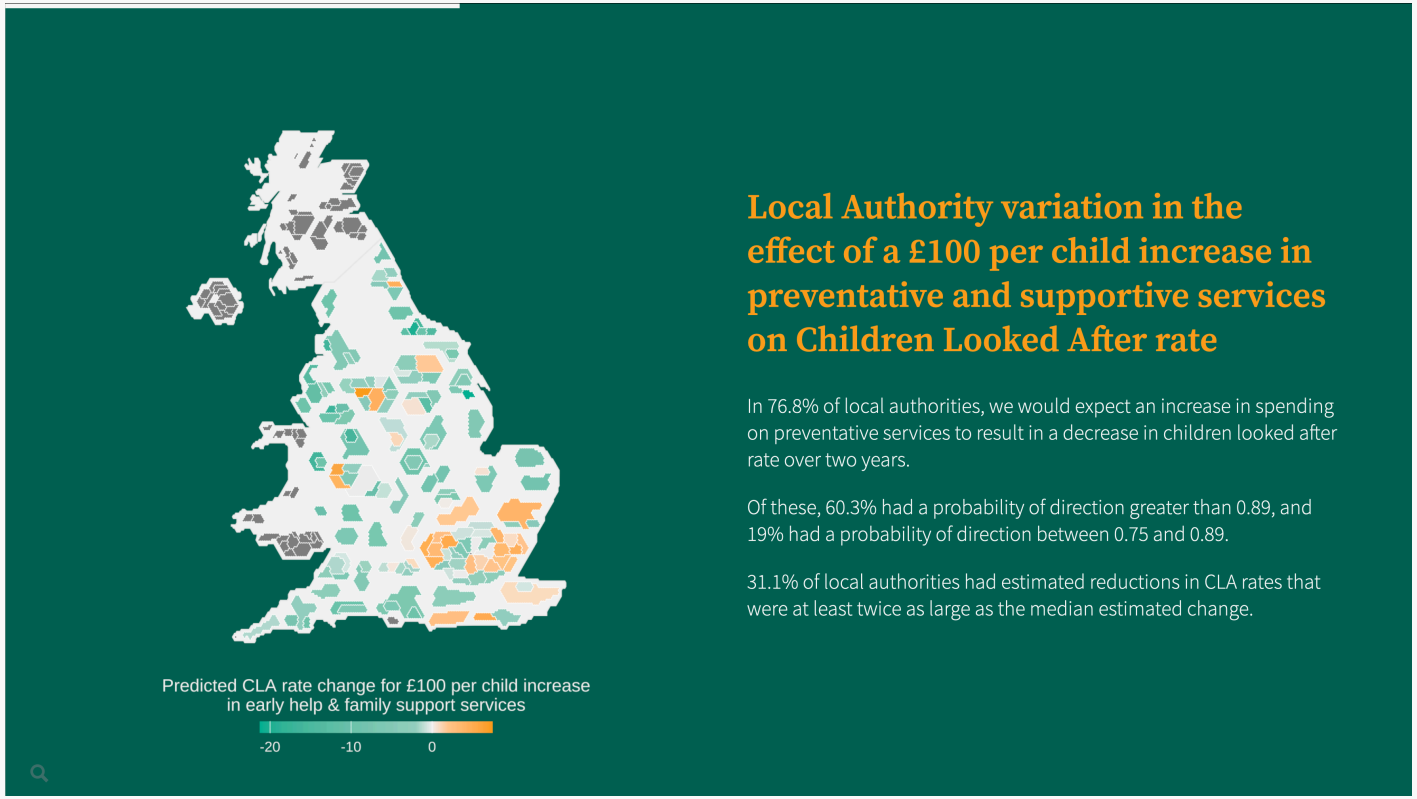
- Here's a map from the same report in the report



Figure 7: The above hex cartogram (developed by the House of Commons Library) shows the degree of variation in the average effect of preventative services spending on children looked after rates between 2009-10 and 2021-22 across local authorities in England.

In some local authorities a cut of £100 per child in preventative spending might be associated with as much as a 15 or 20 per 10,000 increase in children looked after

- Here's a map from the same report in the report
- But here's how it looks on the website on a phone ratio



Local Authority variation in the effect of a £100 per child increase in preventative and supportive services on Children Looked After rate

In 76.8% of local authorities, we would expect an increase in spending on preventative services to result in a decrease in children looked after rate over two years.

Of these, 60.3% had a probability of direction greater than 0.89, and 19% had a probability of direction between 0.75 and 0.89.

31.1% of local authorities had estimated reductions in CLA rates that were at least twice as large as the median estimated change.

- Here's a map from the same report in the report
- But here's how it looks on the website on a phone ratio
- And on my presentation slides

Again, the background changes to make it seamless and in the phone and presentation slide versions I have **removed the place names**, because that information could be hidden and only shown on tap/hover/click. I also cut off

Scotland and Northern Ireland to maximise the size and give the text the best chance of

Publishing a data visualisation: Print

How do we optimise the publication of data visualisation in print?

Things we want to consider:

- Layout, aspect ratio and orientation
- In-line versus stand-alone (splitting up the page)
- Resolution (dpi)
- Colour (or lack thereof!)
- File types (raster or vector images)

All of these sound obvious, but on a big project it can be easy for them to go wrong! **You can have beautiful design, but if it doesn't suit the publication medium, it will look bad!**

How do we optimise the publication of data visualisation in print?

Things we want to consider:

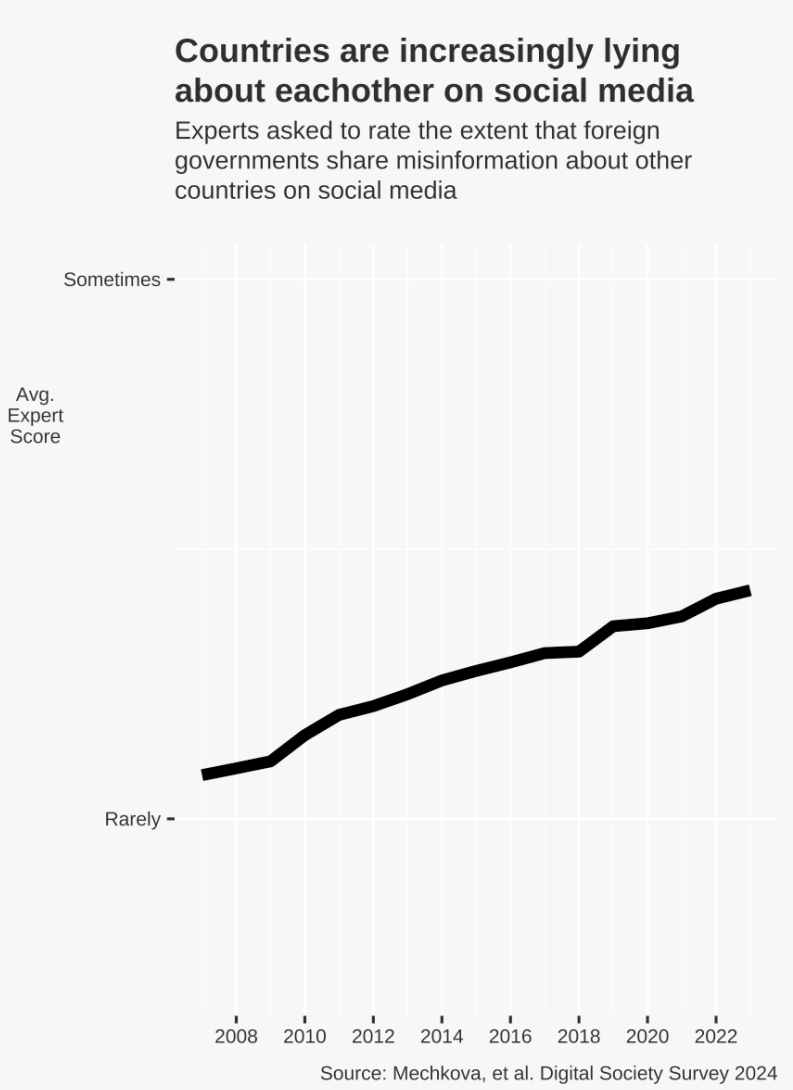
- **Layout, aspect ratio and orientation**
- In-line versus stand-alone (splitting up the page)
- Resolution (dpi)
- Colour (or lack thereof!)
- File types (raster or vector images)

- Portrait
- Landscape

How do we optimise the publication of data visualisation in print?

Things we want to consider:

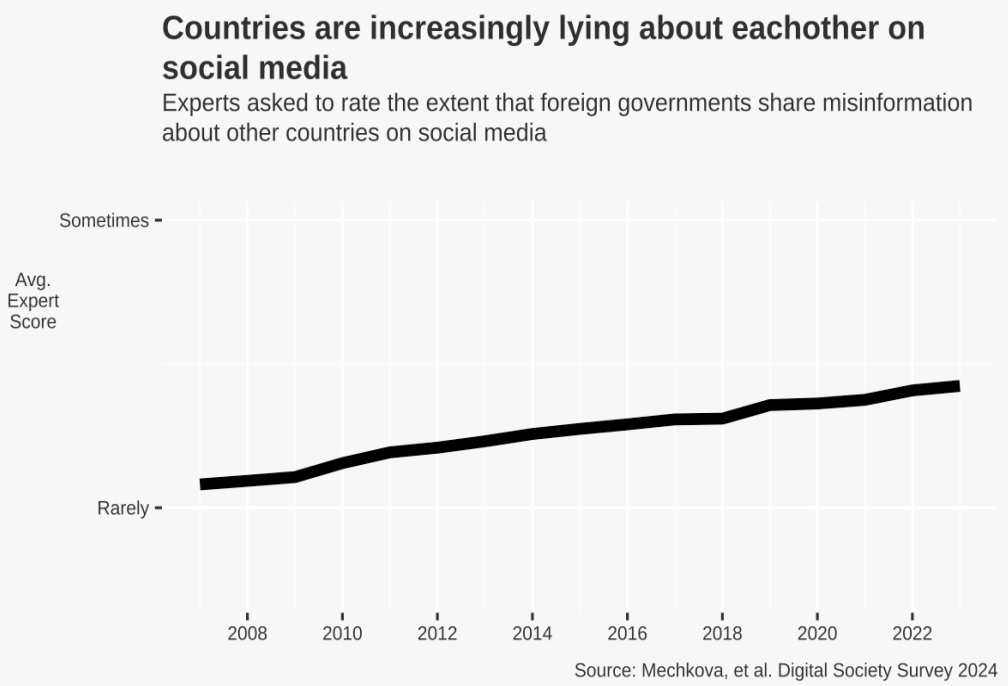
- **Layout, aspect ratio and orientation**
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- Resolution (dpi)
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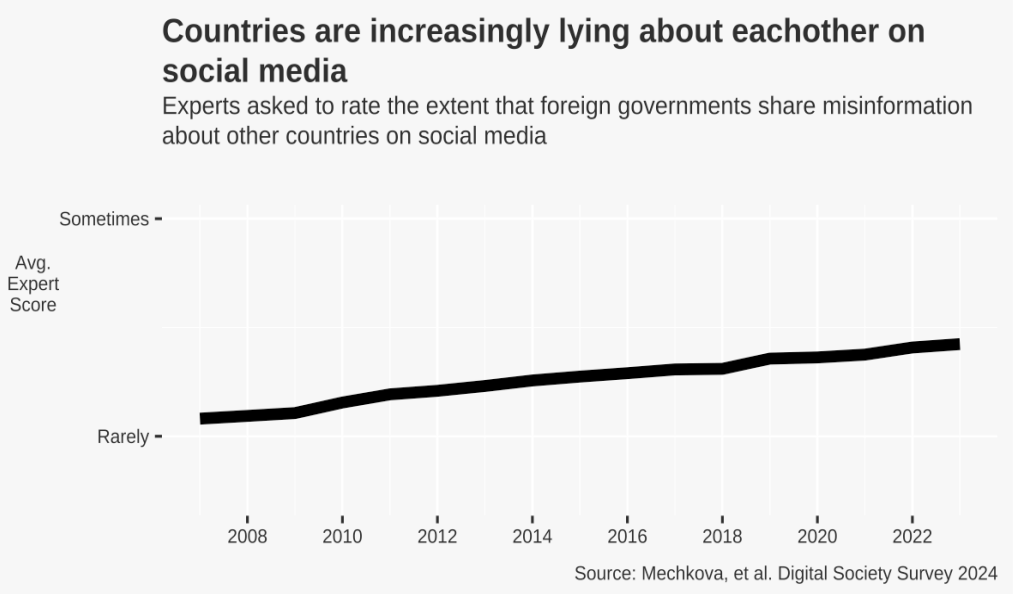


How do we optimise the publication of data visualisation in print?

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- **In-line versus stand-alone (splitting up the page)**
- Resolution (dpi)
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- File types (raster or vector images)

Most attention on the issue has been focused on the spreading of misinformation by unorganised individuals, at the expense of analysing the role that governments play in disseminating misinformation.

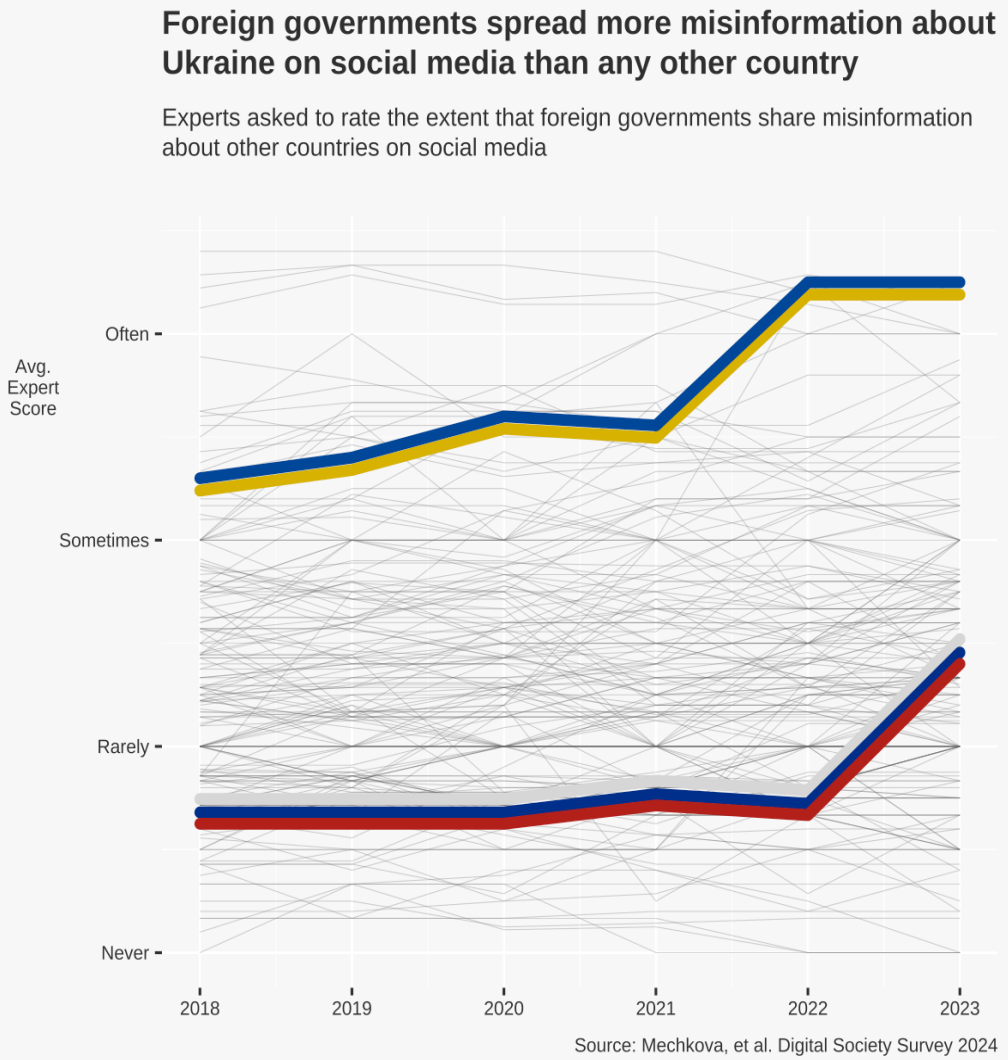


Experts are increasingly recognising that a tactic of foreign governments tends to include disseminating false of misleading claims about rival countries through official agencies on social media. Perhaps because social media is seen as a "less formal" means of communication, where intent can often be hard to discern...

How do we optimise the publication of data visualisation in print?

Things we want to consider:

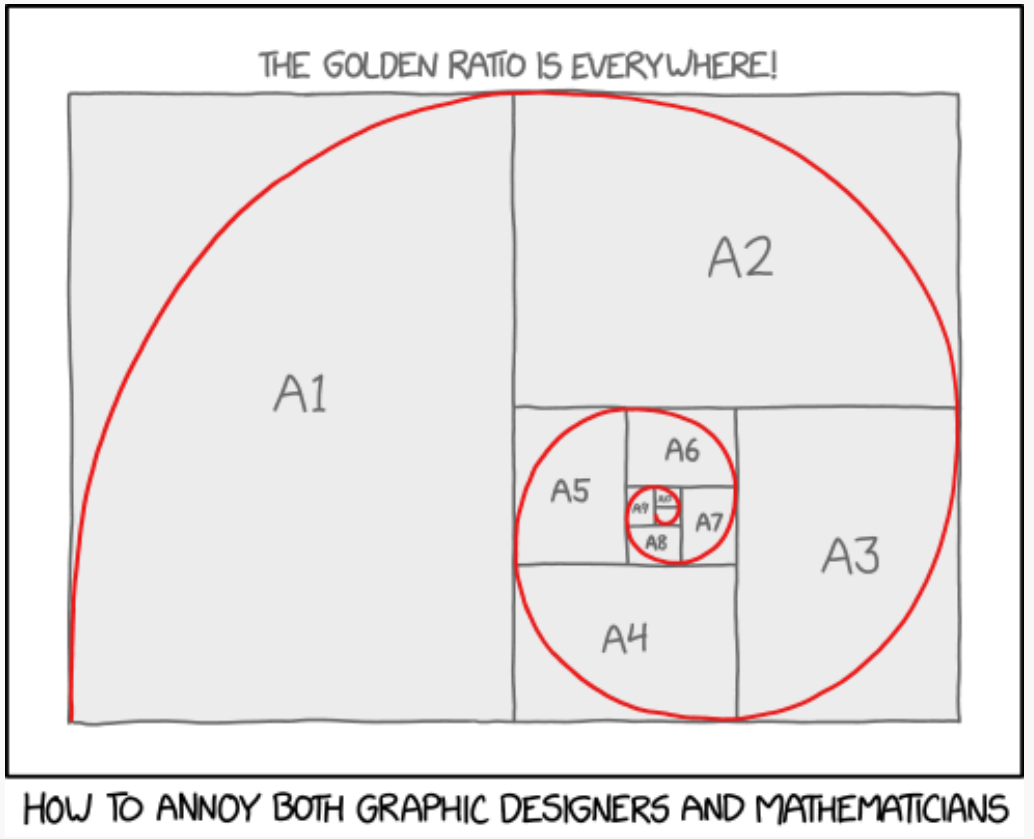
- Layout, aspect ratio and orientation
- **In-line versus stand-alone (splitting up the page)**
  - More complex data visualisations (e.g. facets) showing variation may benefit from being stand-alone on their own page.
- Resolution (dpi)
- Colour (or lack thereof!)
- File types (raster or vector images)



How do we optimise the publication of data visualisation in print?

Things we want to consider:

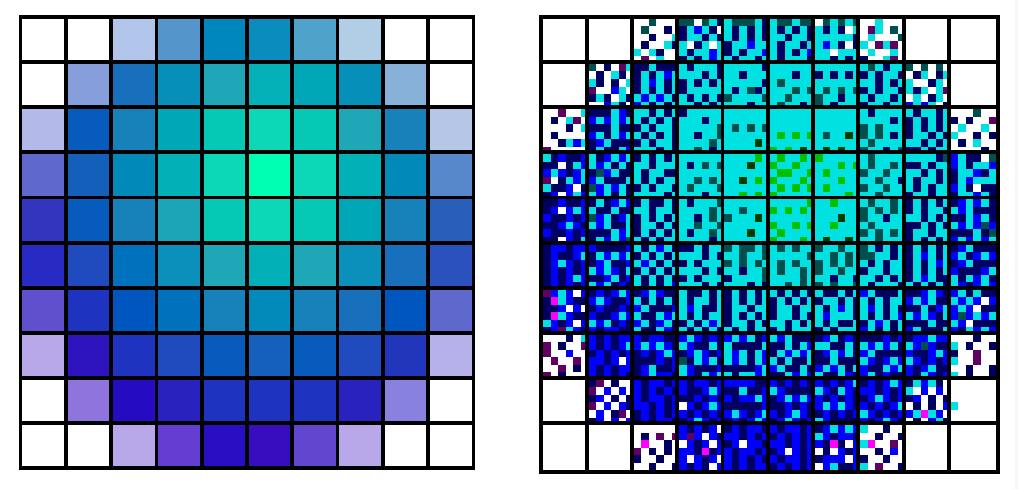
- Layout, aspect ratio and orientation
- **In-line versus stand-alone (splitting up the page)**
  - Think about the most appropriate orientation depending on *how much of the page* your data visualisation is taking up.
- Resolution (dpi)
- Colour (or lack thereof!)
- File types (raster or vector images)



How do we optimise the publication of data visualisation in print?

Things we want to consider:

- Layout, aspect ratio and orientation
- In-line versus stand-alone (splitting up the page)
- **Resolution (dpi)**
  - Refers to the "quality" and detail of the image. On paper there is no loading time, so data visualisations can be as high a resolution (dpi = dots per inch) as the printer allows. Some publishers will have requirements for dpi.
- Colour (or lack thereof!)

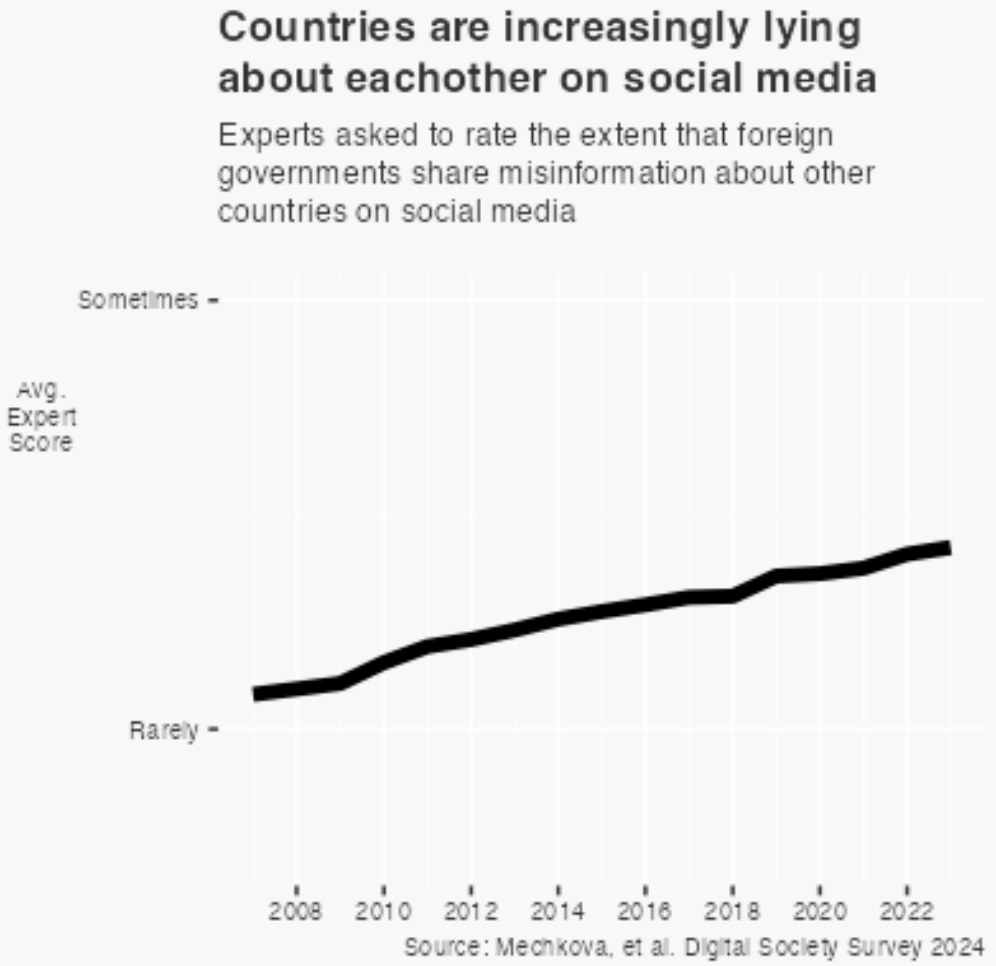


- File types (raster or vector images)

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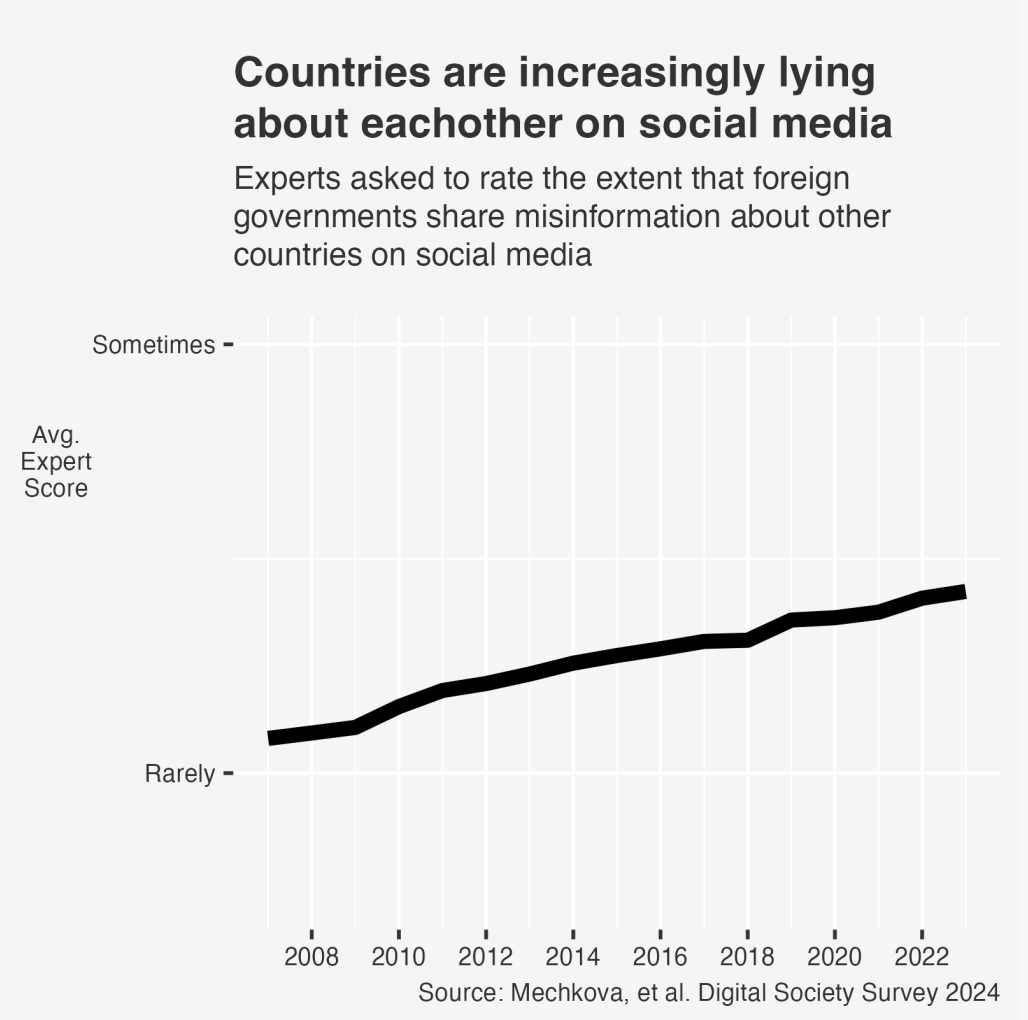
Plot at 100dpi (20kb)

- File types (raster or vector images)

How do we optimise the publication of data visualisation in print?

Things we want to consider:

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- In-line versus stand-alone (splitting up the page)
- **Resolution (dpi)**
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- Colour (or lack thereof!)



Plot at 400dpi (100kb)

- File types (raster or vector images)

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- Colour (or lack thereof!)
- File types (raster or vector images)

> "Line artwork should be supplied in tif or eps format, black and white1 (also known as 1- bit), **resolution 1200 dpi at final size**. Combination artwork (line/tone) should be supplied in tif or eps format, grayscale (also known as 8-bit), **resolution: 800 dpi at final size.**"

Requirements for authors from the Journal of Social Policy (2024)

How do we optimise the publication of data visualisation in print?

Things we want to consider:

- Layout, aspect ratio and orientation
- In-line versus stand-alone (splitting up the page)
- Resolution (dpi)
- **Colour (or lack thereof!)**
  - Does the publication allow colour? If so, how much colour? Colour backgrounds? A limited number of colours?
- File types (raster or vector images)

If colour is allowed:

- High contrast works better (at least 20% difference between colours)
- Avoid white on black for text
- Avoid text font sizes smaller than size 6
- Avoid transparency, use low saturation shades that blend into the background instead
- **Consider the base colour of the paper it's being printed on** — if it's brilliant white, use a more grey font colour (to avoid the contrast being too high). If it's off-white (e.g. newspaper), use black.

How do we optimise the publication of data visualisation in print?

Things we want to consider:

- Layout, aspect ratio and orientation
- In-line versus stand-alone (splitting up the page)
- Resolution (dpi)
- Colour (or lack thereof!)
- **File types (raster or vector images)**
    - Files come in many types: .png, .jpeg/.jpg, .tif/.tiff, .svg, .eps. Some file types are *rasterised* (drawn with pixels/dots), while others are *vectorised* (drawn with maths). Raster files work better when there is a lot of unique elements and colours, vector files work better when

- **.TIFF** files are still popular with print media today because they are lossless (there is no compression to make the file size smaller — remember, we don't need to worry about file size because paper doesn't need to load!) and support more colours (other file formats were designed to be used with the limitations of screens).

Saving as .tiff: `ggsave("ex_plot1.tiff"), width = 4, height = 4)`

- **.EPS** files are the vectorised file of choice for print media. They store instructions on how to reproduce the shapes on your plot by drawing them, which means the data visualisation can be reproduced with no loss in quality at any size (because the shape is not broken down into pixels).

Saving as .eps: `ggsave("ex_plot1.eps"), width = 4, height = 4)`

However, only professionals may have the software required to view these formats. More common formats that everyone can view (e.g. on phones) might be .png or .jpeg

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Another example of a vector and raster image here

Publishing a data visualisation: Web

Things we want to consider:

How do we optimise the publication of data visualisation online?

- **Device (laptop/desktop, mobile, tablet)**
- Common aspect ratios and formats
- In-line versus standalone versus dashboard
- Interactive or static
- File types (raster or vector images)

What kind of device is your data visualisation intended to be viewed on?

- Do you need a different version of your data visualisation for **mobile phones**?
- Do you need a different version of your data visualisation for **social media**?
- Do you need a different version of your data visualisation for **desktop sites**?

More unique versions = more time, but could mean higher engagement and a more professional looking approach.

Things we want to consider:

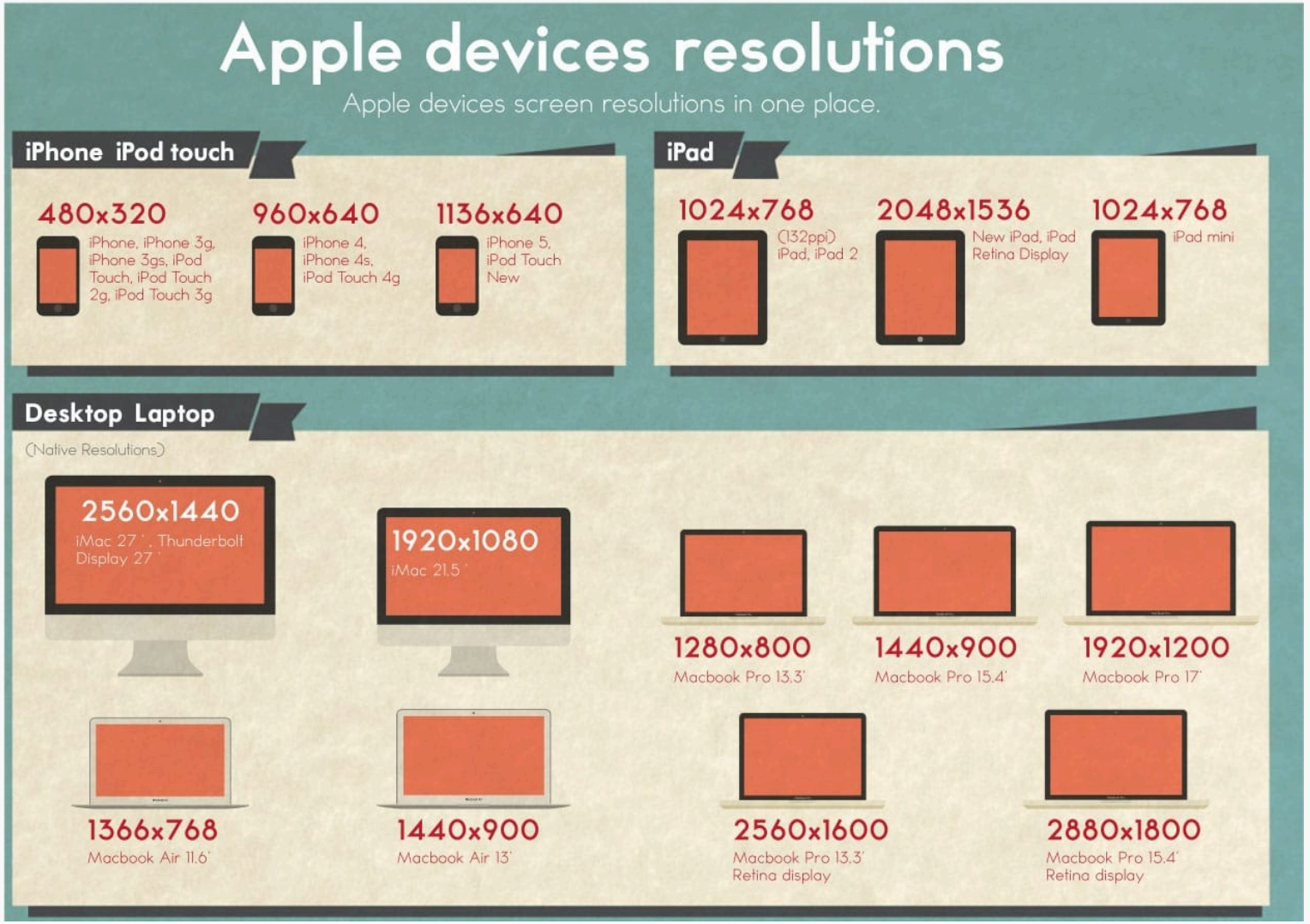
How do we optimise the publication of data visualisation online?

- Device (laptop/desktop, mobile, tablet)
- **Common aspect ratios and formats**
- In-line versus standalone versus dashboard
- Interactive or static
- File types (raster or vector images)

Web images tend to have a fixed value of **72ppi** (pixels per inch) — much lower than the 600dpi of print media.

Rather, increased quality tends to be linked to **increased resolution** of images, for example:

- 1280 pixels (width) x 720 pixels (height), a.k.a. 720p
- 1920 x 1080 a.k.a. 1080p
- 3840 × 2160 a.k.a. 4k Ultra HD


Apple devices resolutions
Apple devices screen resolutions in one place.
iPhone iPod touch
480x320
iPhone, iPhone 3g, iPhone 3gs, iPod Touch, iPod Touch 2g, iPod Touch 3g
960x640
iPhone 4, iPhone 4s, iPod Touch 4g
1136x640
iPhone 5, iPod Touch New
iPad
1024x768
(132ppi) iPad, iPad 2
2048x1536
New iPad, iPad Retina Display
1024x768
iPad mini
Desktop Laptop
(Native Resolutions)
2560x1440
iMac 27', Thunderbolt Display 27'
1920x1080
iMac 21.5'
1280x800
Macbook Pro 13.3'
1440x900
Macbook Pro 15.4'
1920x1200
Macbook Pro 17'
1366x768
Macbook Air 11.6'
1440x900
Macbook Air 13'
2560x1600
Macbook Pro 13.3' Retina display
2880x1800
Macbook Pro 15.4' Retina display

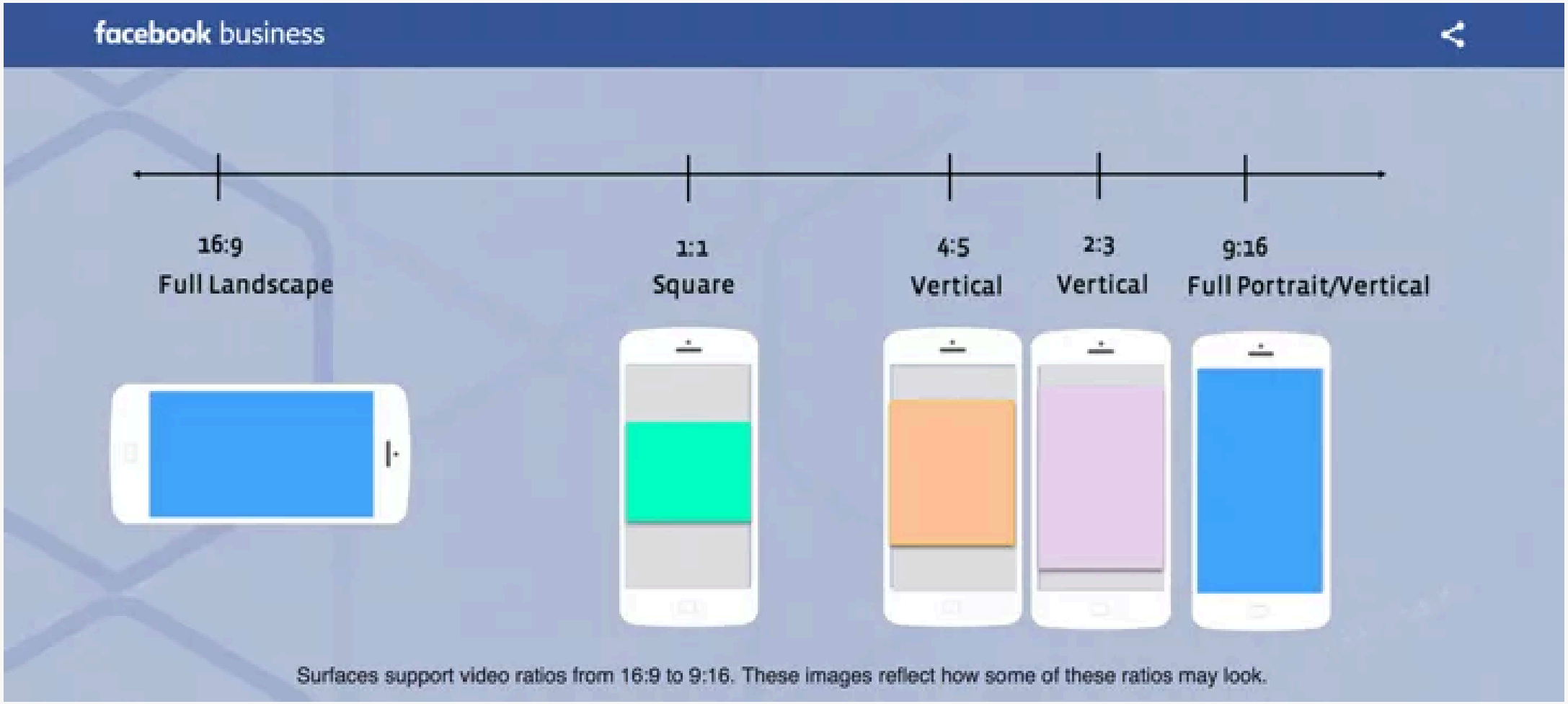
Social Media Image Dimensions 2022

				
Profile photo	320 x 320	360 x 360	200 x 200	400 x 400
Square	1080 x 1080	1200 x 1200	N/A	1200 x 1200
Landscape	1080 x 607	1200 x 630	N/A	1200 x 628
Portrait	1080 x 1350	1200 x 1800	1080 x 1920	628 x 1200
Cover photo	N/A	820 x 312	N/A	1128 x 191

(recommended sizes)



STUDIO ON POINT


facebook business
16:9
Full Landscape
1:1
Square
4:5
Vertical
2:3
Vertical
9:16
Full Portrait/Vertical
Surfaces support video ratios from 16:9 to 9:16. These images reflect how some of these ratios may look.

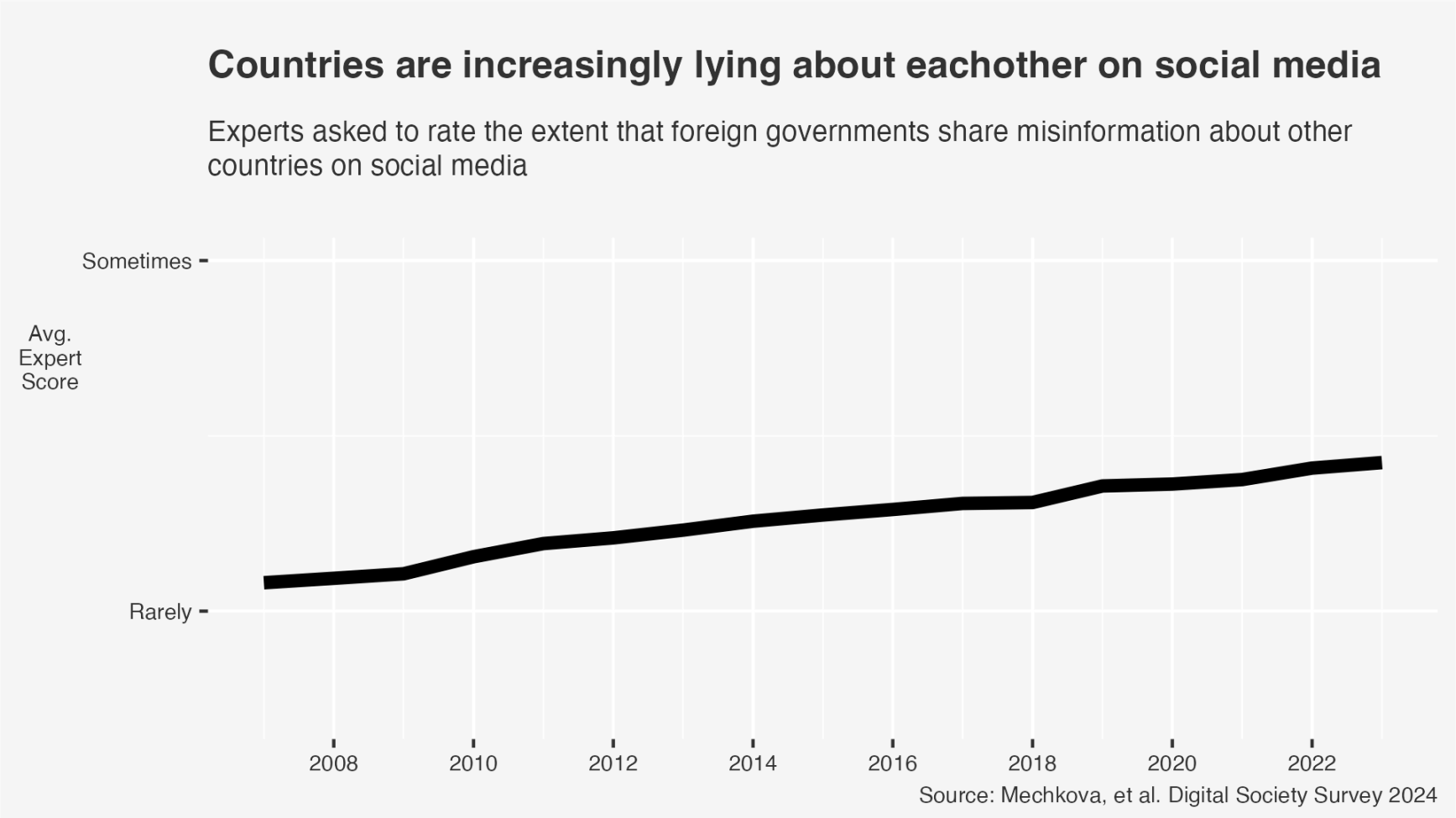
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- In-line versus standalone versus dashboard
- Interactive or static
- File types (raster or vector images)

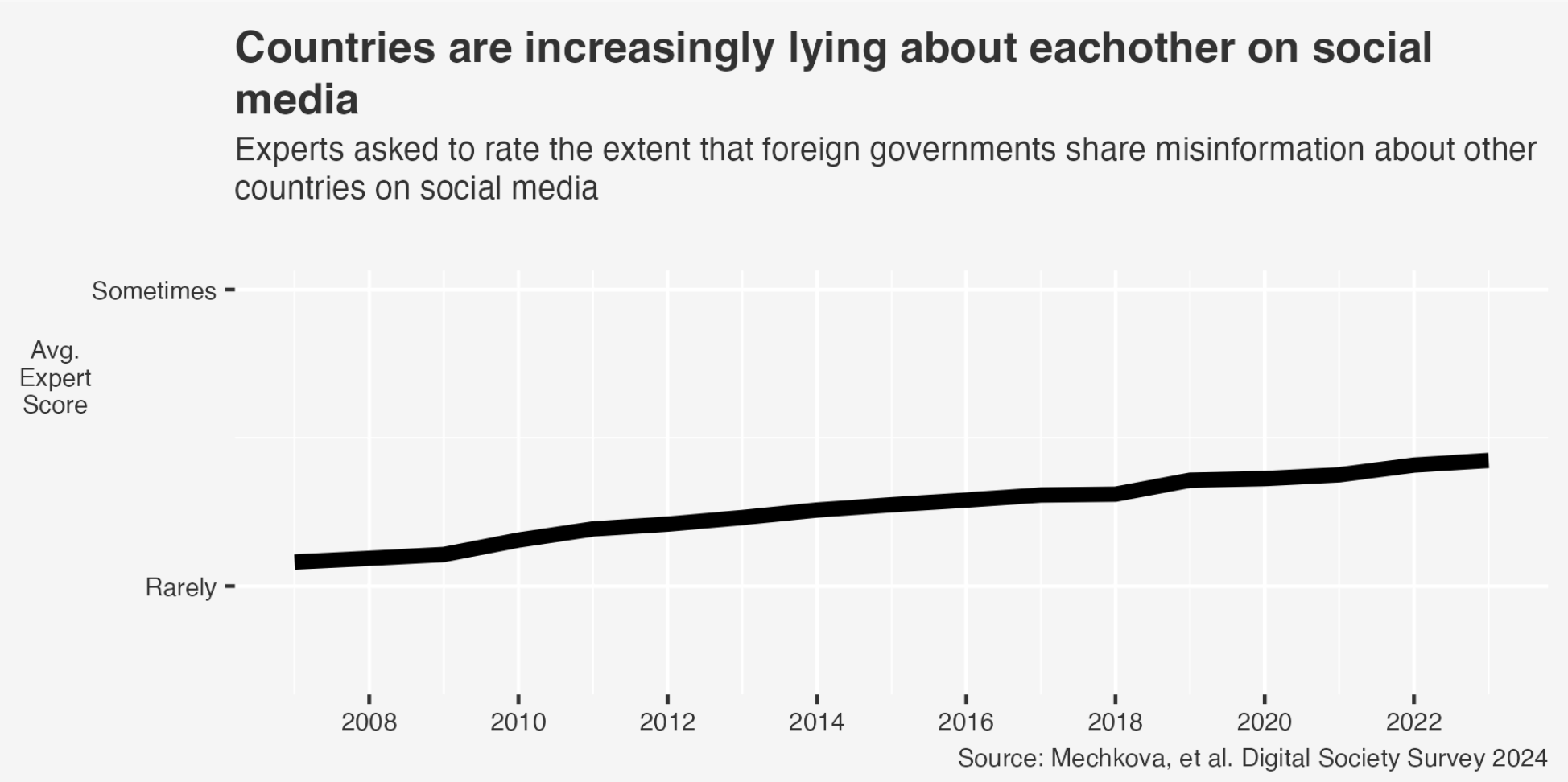
More pixels doesn't always mean better, the best number of pixels is the *minimum number required to provide a clear, crisp graphic*. Too many pixels means the graphic will take longer to load, which is problematic because:

- The probability of someone giving up on the image loading increases by 32% between 1 second and 3 seconds.
- Your graphic will take longer to load with excess resolution/file size in countries with poorer internet access
- You are trading a clear image for a *bad user experience*.



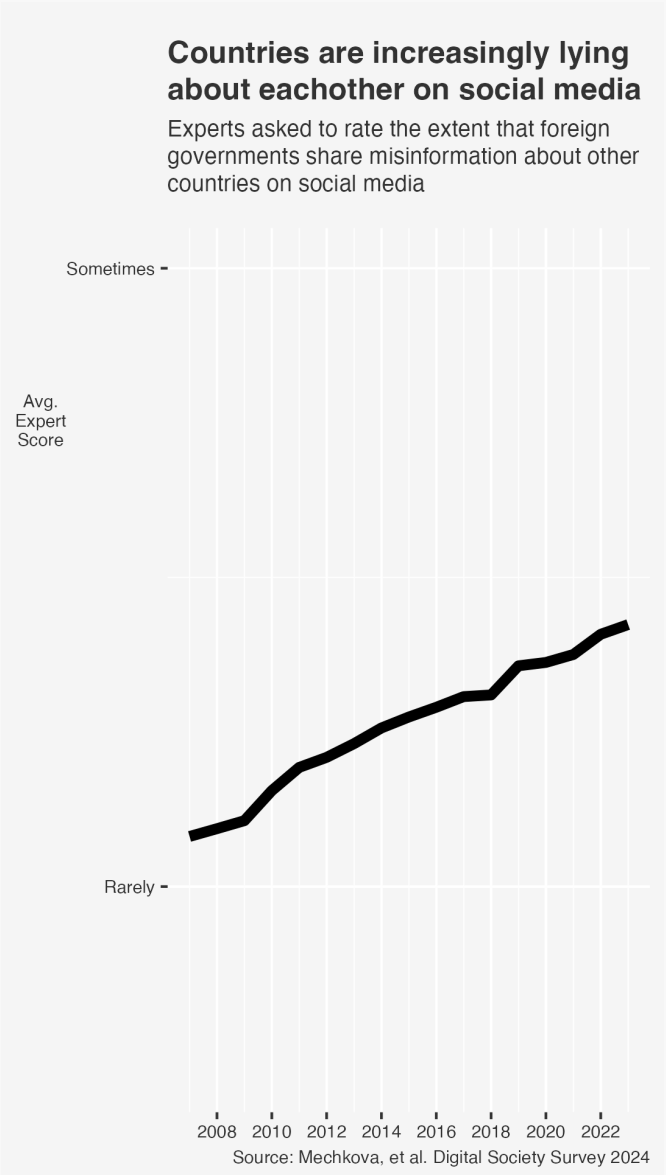

```
ggsave(filename = "ex_plot1_1080p.png", width = 1920, height = 1080, units = "px")
```

1920 x 1080 (Full HD), 16:9 Aspect Ratio




```
ggsave(filename = "ex_plot1_twitter.png", width = 1400, height = 700, units = "px", scale = 1.3)
```

1400 x 700 (Twitter/Facebook), 2:1 Aspect Ratio




```
ggsave(filename = "ex_plot1_mobileportrait.png",
width = 640, height = 1136, units = "px", scale =
1.7)
```

640 x 1136 (Phone Portrait), 9:16 Aspect Ratio

Things we want to consider:

How do we optimise the publication of data visualisation online?

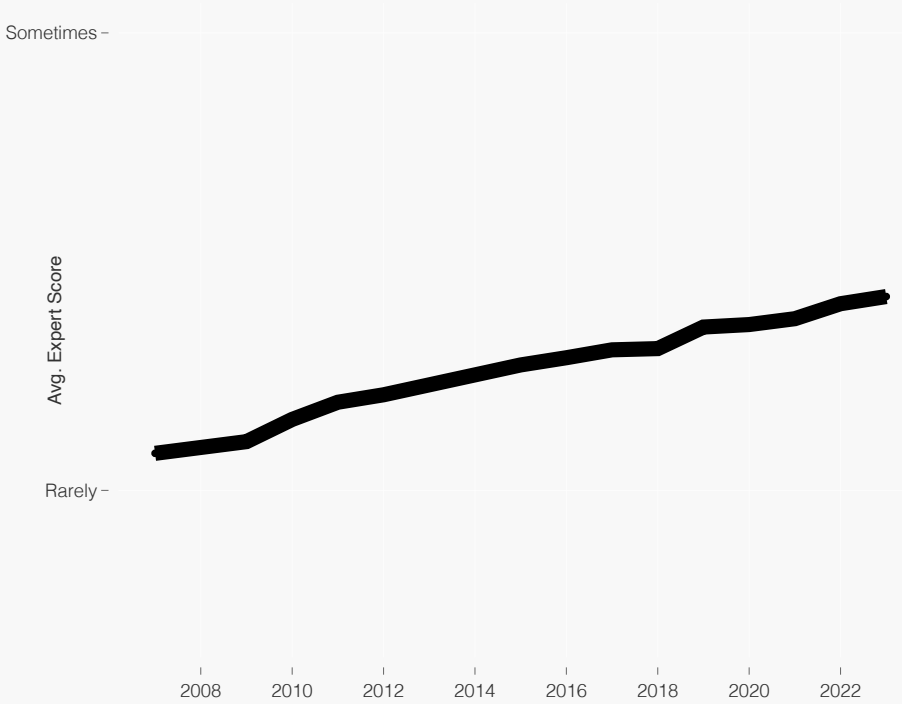
- Device (laptop/desktop, mobile, tablet)
- Common aspect ratios and formats
- **In-line versus standalone versus dashboard**
- Interactive or static
- File types (raster or vector images)

Things we want to consider:

How do we optimise the publication of data visualisation online?

- Device (laptop/desktop, mobile, tablet)
- Common aspect ratios and formats
- In-line versus standalone versus dashboard
- **Interactive or static**
    - Web platforms allow for interactivity (additional information to be encoded into the data visualisation and displayed on click or hover)
    - This also includes reactively re-sizing to the screen size — a nice benefit of interactive visualisations!

Countries are increasingly lying about eachother on social media



Things we want to consider:

How do we optimise the publication of data visualisation online?

- Device (laptop/desktop, mobile, tablet)
- Common aspect ratios and formats
- In-line versus standalone versus dashboard
- Interactive or static
- **File types (raster or vector images)**

.PNG files are the most common rasterised file format for web graphics (and for a lot of print). They support transparency and usually do not lose as much quality when compressed (file size is reduced to save space).

```
ggsave("ex_plot1.png")
```

.JPG/JPEG files are also a common rasterised file format, but is now less common than png files.

```
ggsave("ex_plot1.jpg")
```

.SVG files are the most common vector file format found online, commonly used for text logos and infographics. Like other vectore file formats, SVG files do not support transparency.

```
ggsave("ex_plot1.svg")
```

Summary

- While the idea of taking your data visualisation from R and into publication after applying good graphic design principles may seem straightforward, there are a lot of considerations to make to ensure your data visualisation works *within* a publication.
- There are some common formats and standards that you can follow in order to identify how to size, set the quality, and include interaction or other features within your data visualisation.
- Key questions to ask yourself:
    - Print? Digital? Both?
    - Is there a one-size fits all for your audiences, or do you need some different versions?
    - How can you balance visual quality and user experience (speed of loading)?
    - Should your visualisation be viewed in-line with text, or is it standalone?

Workshop this week

Let's start drawing **maps**. We're going to work with some existing data that includes spatial information to draw different kinds of maps, and consider the information that needs to be included when we draw maps.

Tasks to look at in advance of Week 8

Core tasks:

Before lecture: Find a map in the news that is being used to present some data. Then, share on Blackboard: (i) the map you find, (ii) where it was published (if that's not obvious) and (iii) one observation about how well, or not, you think it is integrated into the place where it has been published (one sentence will do).

Before lecture: Read When maps shouldn't be maps http://www.ericson.net/content/2011/10/when-maps-shouldnt-be-maps/

Before workshop: Work through the task at the end of the workshop handout

Supplementary tasks:

- Before workshop: Read chapter 7/Draw maps of the Healy book